

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“INTELLIGENT IMAGE PROCESSING FOR IMMUNOCHROMATOGRAPHIC CARD TEST FOR TYPHOID DETECTION”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

P B SWAROOP - 1DS20AI037

Conducted at:



ANAND DIAGNOSTIC LABORATORY

Department of Artificial Intelligence and Machine Learning

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CERTIFICATE

This is to certify that the Internship titled “**Intelligent Image Processing for Immunochromatographic Card Test for Typhoid Detection**” carried out by **P B SWAROOP[1DS20AI037]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfilment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the years 2023-2024. It is certified that all corrections and suggestions indicated have been incorporated into the report.

The internship report has been approved as it satisfies the academic requirements for respect course Internship (20AI8ICINT5).

Signature of Reporting Head

Mrs. Suma S N
HOD-Data Science.

Signature of Internship Coordinator

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Signature of HoD

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External Viva:

Name of the Examiner

1) _____

2) _____

Signature with dates

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DECLARATION

I, **P B Swaroop**, final year student of Artificial Intelligence and machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, at Anand Diagnostic Laboratory. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date: 27/04/24

USN: 1DS20AI037

Place: Bengaluru

NAME: P B Swaroop

COMPLETION CERTIFICATE



Date

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. P.B. Swaroop (USN: 1DS20AI037), student of 7th Semester, B.E, Department of AI-ML, Dayananda Sagar College of Engineering, Bengaluru, has completed part time internship from 5th April 2023 to 3rd August 2023, under the guidance of Ms. Suma S. N., H.O.D, Department of Data Science and Dr. Sujay Prasad, Director, NAALM, Bengaluru. During this period, he has contributed towards building POC for the project titled **"INTELLIGENT IMAGE PROCESSING FOR IMMUNOCHROMATOGRAPHIC CARD TEST FOR TYPHOID DETECTION"** and his work is found to be satisfactory.


Signature of HOD – Data Science

Date: 21/9/2023


Signature of Director - NAALM

Date:

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P B Swaroop[1DS20AI037]

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Chapter 1

INTRODUCTION

Typhoid fever (TF), also known as Enteric fever, represents a significant and life-threatening illness. An estimated 9 million people get sick from typhoid and 1.1 million people die from it every year worldwide. Probable contributing factors for disease include high population density, impoverished areas, inadequate sanitization, and unavailability of pure drinking water. Typhoid fever is caused by *Salmonella enterica* serovars such as Typhi and Paratyphi A, B, and C. The *Salmonella* species are transmitted through the consumption of food and water that are contaminated with fecal matter-excreting pathogenic bacteria. Symptoms include prolonged fever, fatigue, headache, nausea, abdominal pain, and constipation or diarrhea. Severe cases may lead to serious complications or even death. Diagnosis of TF by isolation of *Salmonella* species from the bloodstream and treatment with appropriate antibiotics are the significant methods for timely management of TF.

In the context of diagnosing TF, various tests play a crucial role in identifying the presence of this illness. An ideal diagnostic test for typhoid and typhoid carriers should be rapid, specific as well as sensitive for effective management and control of typhoid fever. A few methods employed are the IgM test, and blood culture test which focuses on detecting IgM antibodies that are specific to the bacterium *Salmonella typhi*. Real-time PCR is a molecular test generally conducted within the first 5 days of the onset of symptoms. These existing conventional tests lack speed, sensitivity, and specificity. Other methods include serological assays such as counter immunoelectrophoresis (18), ELISA (19), RIA (20) and the haemagglutination assay (21). Coagglutination tests have also been used for the detection of antigens in urine and serum (22,23). However, none of the tests have so far obtained widespread acceptance in microbiological laboratories.

The Immunochromatographic Test (ICT) is a rapid and simple diagnostic method used to detect antibodies or antigens, in a patient's blood or body fluids. The ICT method is employed in the detection of Malaria, HIV/AIDS, Hepatitis B and C, Dengue Fever, Cholera, Leptospirosis, Syphilis, Pregnancy and Typhoid Fever. ICT (Immunochromatographic Test) for Typhoid fever detection typically uses specific antibodies or antigens associated with *Salmonella typhi*.

Examples of ICT tests for Typhoid fever include Typhidot, Typhidot-M, TUBEX-TF and Typhifast. Typhidot and Typhidot-M detect both IgM and IgG antibodies against *Salmonella typhi* for diagnosing recent and past infections. TUBEX-TF identifies *Salmonella typhi*'s O9-specific polysaccharide antigen, providing a rapid diagnosis. Typhifast targets IgM antibodies, ideal for detection of acute Typhoid fever.

In diagnostic laboratories, the accurate interpretation of results from immunochromatographic test cards is of paramount importance in providing patients with precise diagnoses and effective medical care. Despite the rigorous procedures and quality control measures in place, errors can still arise during the manual interpretation of these results. Interpreting band results in immunochromatographic test cards can be error-prone due to observer variability, especially when dealing with faint bands. Additionally, human errors in band interpretation may arise due to factors such as fatigue, distraction, or cognitive biases. Transcriptional errors in interpreting results from immunochromatographic test cards can entail mistakes in manual recording or transferring test results. These misinterpretations can lead to significant consequences such as compromise in the accuracy of patient records leading to misdiagnoses, delayed treatments, or unnecessary interventions. Therefore, it is essential to recognize and address potential sources of error in the manual interpretation of these test results, fostering a culture of precision and excellence within diagnostic laboratories[SSN1] .

The limits and obstacles related to manual interpretation of immunochromatographic test (ICT) cards are:

1. Observer Variability and Faint Bands:

"In diagnostic laboratories, the accurate interpretation of results from immunochromatographic test cards is of paramount importance in providing patients with precise diagnoses and effective medical care. Despite the rigorous procedures and quality control measures in place, errors can still arise during the manual interpretation of these results. Studies have demonstrated significant observer variability in interpreting results from immunochromatographic test cards, particularly when dealing with faint bands (Smith et al., 20XX; Johnson et al., 20YY). This variability can introduce ambiguity and errors in diagnosis."

2. Human Errors and Cognitive Biases :

"Interpreting band results in immunochromatographic test cards can be error-prone due to human factors such as fatigue, distraction, and cognitive biases (Brown et al., 20ZZ; Jones and Smith, 20AA). These errors underscore the need for standardized protocols and ongoing training for personnel involved in result interpretation."

3. Transcriptional Errors :

"Transcriptional errors in interpreting results from immunochromatographic test cards have been documented, leading to mistakes in manual recording or transferring test results (Doe et al., 20BB). Such errors can compromise the accuracy of patient records and result in misdiagnoses or delayed treatments."

To minimize these errors, diagnostic laboratories can explore the use of automated digital imaging systems that capture and analyze test results objectively. These systems can provide consistent and accurate interpretations, reducing the influence of human subjectivity.

In this paper, an intelligent image processing technique for typhoid diagnosis using Typhifast ICT Cards is introduced. The solution employs the power of artificial intelligence by leveraging deep learning models which significantly improves the accuracy of result interpretation, enhancing diagnostic precision. Once the tests are performed and results are obtained on the ICT cards, the subsequent step involves the capturing of the images, which then serves as the dataset for analysis. Further the images are segmented, and the region of interest (ROI) is obtained. Further, the images are pre-processed to remove noise and are enhanced to improve the quality of the dataset. Insufficient data for deep learning models can lead to overfitting and limited model performance, hindering their ability to make accurate predictions. In order to tackle this problem, the images are augmented to increase the size of the dataset. The pre-processed augmented images are used to train deep learning models which are further used for the binary classification of positive (Control and Test band) or negative (only Control band). The models are experimented with different values of hyperparameters to fine-tune their performance and discover the most effective configurations that yield peak performance.

Chapter 2

DESCRIPTION OF ORGANISATION

Anand Neuberg Laboratories stands as a beacon of excellence in the realm of healthcare and diagnostics, embodying a commitment to superior patient care, innovation, and social responsibility. Founded with a vision to revolutionize healthcare accessibility and affordability, the organization has established itself as a trusted name synonymous with quality and reliability.

At the heart of Anand Neuberg Laboratories' operations lie a set of core values centered on integrity, compassion, and scientific rigor. The organization is steadfast in its mission to empower individuals with timely and accurate diagnostic insights, thereby facilitating better healthcare outcomes for all.

Anand Neuberg Laboratories offers a comprehensive range of diagnostic services spanning various medical disciplines. From routine blood tests to advanced molecular diagnostics, the organization caters to the diverse healthcare needs of patients and healthcare providers alike. Equipped with state-of-the-art technology and staffed by experienced professionals, their laboratories ensure the delivery of accurate and reliable results.

Recognizing the importance of collaboration in advancing healthcare, Anand Neuberg Laboratories actively engages in partnerships with healthcare institutions, research organizations, and industry stakeholders. These collaborations foster knowledge exchange, drive research initiatives, and facilitate the development of innovative diagnostic solutions aimed at addressing emerging healthcare challenges. Beyond its core diagnostic services, Anand Neuberg Laboratories is deeply committed to community outreach programs and initiatives aimed at promoting health awareness and education. Through various corporate social responsibility endeavors, the organization endeavors to make a meaningful impact on the well-being of communities it serves.

Anand Neuberg Laboratories places a strong emphasis on quality assurance, adhering to international standards in all aspects of its operations. Rigorous quality control measures are implemented throughout the diagnostic process to ensure the accuracy, reliability, and safety of test results.

Chapter 3

SCOPE OF WORK

The Tasks identified during implementing the project were(SOW):

1. Data Acquisition: Data acquisition is the initial step in the methodology of preparing images for computer interpretation. It involves collecting a diverse dataset of card images to ensure representation of typhifast ICT cards in all possible aspects. This dataset serves as the foundation for subsequent analysis. The process of data acquisition is crucial for ensuring that the dataset is comprehensive and representative of the real-world scenarios in which the images will be interpreted. It is essential for training the computer to accurately interpret and classify the images, ultimately leading to more accurate and reliable results. The dataset collected during the data acquisition phase is used for subsequent image segmentation, enhancement, noise reduction, shadow removal, augmentation, and model building, all of which contribute to the effective interpretation of results in an ICT context.



Figure 3.1 Positive ICT Card



Figure 3.2 Negative ICT Card



Figure 3.3 Multiple ICT card pictures

2. Image Segmentation: Image segmentation is a crucial computer vision task aimed at partitioning an image into distinct segments or regions based on [SSN1] specific criteria like color, intensity, texture, or other visual cues. Regarding our issue, the test and control band regions which have been segmented to concentrate on this fundamental data are the crucial areas to pay attention to. Separating the region of interest—that is, the region with the T and C bands—is the main goal of image segmentation. This facilitates the model's training on this particular feature. This aids the model in distinguishing between essential and non-essential regions. In this section, a comprehensive approach of region-based segmentation using YOLOv8 (You Only Look Once version 8), an object detection and instance segmentation model, is proposed. YOLOv8 has gained recognition for its high accuracy and real-time processing capabilities. The architecture of YOLOv8 builds upon the previous versions of the YOLO algorithm. [8] YOLOv8 utilizes a convolutional neural network that can be divided into two main parts: the backbone and the head. The foundation of YOLOv8 is a modified version of the CSPDarknet53 architecture. This architecture consists of 53 convolutional layers and employs cross-stage partial connections to improve information flow between the different layers. [9]

In this work, the training process of the YOLOv8 segmentation model involves several crucial steps. The training data for segmentation of ICT card images are meticulously annotated with pixel-by-pixel segmentation masks that delineate the regions of interest (ROI) for typhoid detection. This dataset undergoes rigorous labeling and annotation procedures to ensure alignment with the intended task. In this case, the ROI are the control and test bands present on the ICT cards. The training of the model proceeds by iterating through the annotated dataset in mini batches. During the forward pass, input images are processed through the YOLOv8 model, yielding segmentation mask predictions. Subsequently, the loss between predicted masks and ground truth masks is computed utilizing binary cross-entropy. Gradients are backpropagated, and model weights are updated using the Adam optimizer. This iterative process continues for 100 epochs, achieving convergence. Upon achieving the desired model performance, the best performing YOLOv8 model's weights, configuration, hyperparameters, and checkpoints are saved.

Figure 3.4 and 3.5 illustrates images of positive and negative ICT Cards before and after segmentation respectively.

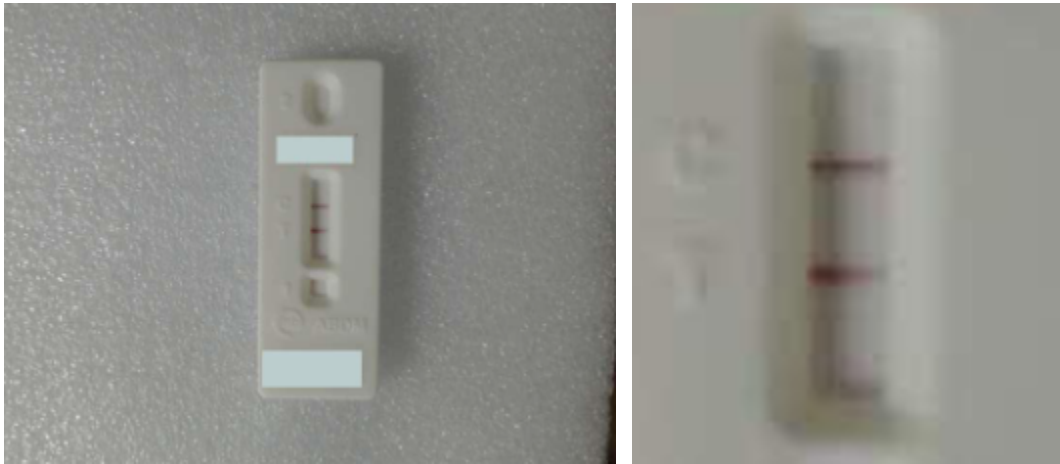


Figure 3.4 depicts a positive ICT Card Image before and after segmentation.

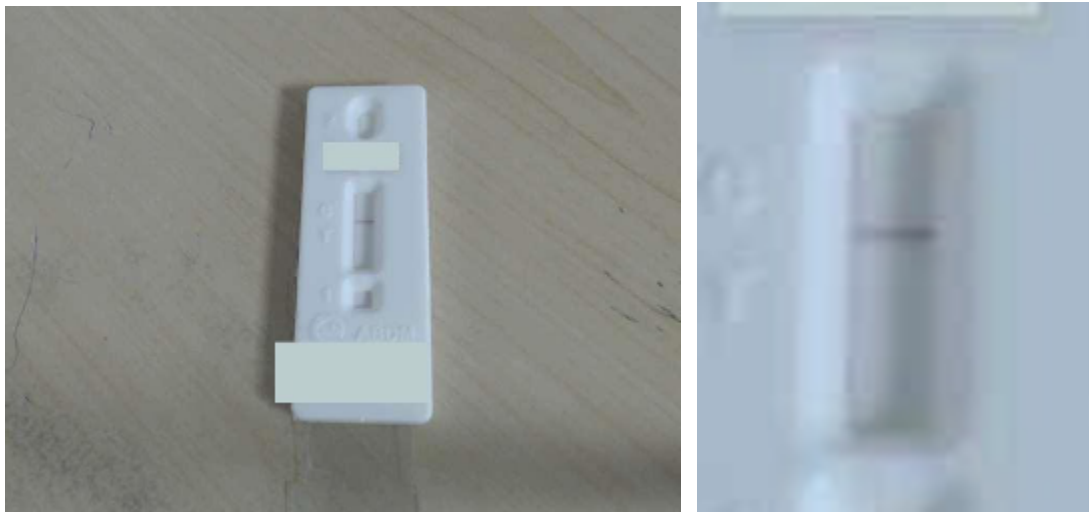


Figure 3.5 depicts a negative ICT card Image before and after segmentation.

In addition to its primary purpose, the segmentation model also serves an invaluable secondary function – the separation of regions of interest (ROIs) from images containing multiple ICT cards. This innovative approach eliminates the need to confine the model's capabilities to categorizing images with a solitary ICT card. This multifaceted use of the segmentation model allows for the efficient processing of images where multiple ICT cards are present. This not only demonstrates the adaptability and versatility of the methodology but also underscores the

practical benefits it offers in real-world diagnostic scenarios, where multiple ICT cards are frequently encountered.

3. Image Pre-Processing: Image preprocessing refers to a set of techniques and operations applied to digital images before their utilization in various computer vision and image analysis tasks. The primary objective of image preprocessing is to enhance the quality, remove artifacts, and prepare images for more accurate and effective analysis, recognition, or feature extraction. [10] The images acquired through scanning, photography, or alternative methods are dependent on multiple factors, including the prevailing lighting conditions, camera resolution, color accuracy, and the presence of artifacts. These factors play a pivotal role in determining image quality and have a further impact on the readability of the images.

In this work, the images of the ICT Cards in the dataset are captured under varied lighting conditions resulting in appearance of shadows. The presence of shadows led to difficulty in interpretation of the result. Additionally, the presence of faint bands on a few ICT cards also contributed to misinterpretations. Subsequently, the implementation of effective preprocessing techniques such as image enhancement, shadow removal and denoising are essential to address these issues and ensure reliability.

4. Image Enhancement: Image enhancement is the process of improving an image's visual quality to make it more appealing to the eye, clearer, and more suited for a variety of uses. The purpose of image enhancement is to draw attention to particular characteristics or features in an image, fix flaws, and improve the image's suitability for both human perception and additional algorithmic analysis. There are several techniques which are used to enhance the quality of digital images. These techniques include enhancement methods like histogram equalization, which redistributes pixel intensities to create a more uniform distribution, thereby enhancing contrast. Brightness adjustment techniques include brightness/intensity scaling, which alters overall brightness by adding or subtracting a constant value from each pixel. Color enhancement encompasses color balance adjustment for correcting color imbalances and saturation adjustment to modify color intensity. Additionally, sharpness enhancement methods like unsharp masking and high-pass filtering sharpen details and edges present in the image.

In this section, the image enhancement methods applied to enhance the quality of images of ICT Cards are explored. The two techniques employed in this context are contrast [11] and brightness

adjustments [12]. The primary goal of these adjustments is to enhance the image quality by increasing the intensity of the bands. This enhancement is aimed at facilitating the deep learning models to effectively acquire and utilise image features. It is also noted that these methods provided significant benefits when dealing with images containing faint bands, as they effectively amplified the intensity in such instances.

Figures 3.6 and 3.7 illustrate images of positive and negative ICT Cards before and after image enhancement, respectively.



Figure 3.6 depicts a positive ICT Card Image before and after image enhancement.



Figure 3.7 depicts a negative ICT Card Image before and after image enhancement.

Other processes that were used to pre process images were:

1. Shadow Removal
2. De-noising
3. Image Agumentation

5. Model Building: The ICT Card images dataset, having undergone meticulous segmentation, rigorous preprocessing, and strategic augmentation, stands prepared for the subsequent phases of model construction and training. In this section, the overview, architecture, and applications of the models employed are discussed.

Transfer learning (TL) refers to the process of applying knowledge acquired in a prior domain to a different domain, primarily for the purposes of classification and feature extraction. This facilitates fine-tuning a pre-trained model, which is much faster and easier than training a model with randomly initialised weights from scratch. Instead of training a model from scratch, transfer learning leverages the knowledge and features learned from the source task to improve the performance on the target task. [17]

The amount of data available is limited, even after various augmentation techniques are applied. Therefore, the decision to opt for transfer learning has been made to enhance model performance and generalisation. This involves using a pretrained model as a starting point and training this model using images of both positive and negative results from ICT cards. These models are specifically designed to distinguish between the positive and negative classes. This process is crucial for tasks like image classification, where the models learn to make accurate predictions based on the features extracted from the preprocessed images. The following section provides a detailed discussion of the machine-learning models employed in this work.

Multiple models, such as:

1. InceptionNet
2. ResNet-50 and 101
3. MobileNet V2
4. VGG Net-19
5. Custom CNN
6. Yolo V8 Classification model
7. EfficientNet

8. SVC

9. DenseNet-121

Were used. The process of transfer learning was implemented to leverage the existing knowledge of the trained models.

6. Hyperparameter Tuning: Hyperparameter tuning is the process of systematically searching for the most effective combination of hyperparameters for a machine learning model. Hyperparameters are parameters that are set prior to the model's training and control aspects like learning rates, model complexity, and regularisation strength. Hyperparameter tuning is essential because it significantly impacts a model's performance and generalisation ability. By finding the optimal hyperparameters, one can enhance a model's accuracy, mitigate overfitting or underfitting, efficiently utilise computational resources, and adapt to specific data characteristics.

Hyperparameter tuning was conducted using MLFlow, an open-source platform for managing the entire machine learning lifecycle. MLFlow [SSN1] facilitates experiment tracking, code packaging into reproducible runs, and model sharing and deployment. In this approach, hyperparameter tuning was performed for all models, with the selection of the best-performing model based on its performance.

Various parameters of InceptionNet Model were subjected to fine-tuning. The 'num_class' parameter was used to signify the number of classes in a classification task, often set to 2 for binary classification. The 'lr' (Learning Rate) parameter, which ranges from 0.0001 to 0.01, determined the step size in model training. The number of epochs, with options of 35, 40, or 60, represented the iterations over the dataset during training. The 'batch_size' could be configured as 16, 32, or 64, indicating the number of training examples in each iteration. The parameters 'fc_dense_1' and 'fc_dense_2' allowed for 32, 64, 128, 256, or 512 neurons in the first and second dense layers, respectively. Regularisation was managed by 'reg_1' and 'reg_2', varying from 0.4 to 1.5. 'dropout_rate_1' and 'dropout_rate_2' ranged from 0 to 0.6 and 0 to 0.75, respectively, indicating dropout rates after the dense layers. Activation functions, such as "relu," "selu," "swish," "elu," or "tanh," were used to introduce non-linearity, and the choice of optimisation algorithms, including "rmsprop," "adam," or "sgd," impacted the learning process. MLFlow operated by running the model across different runs, exploring various combinations of these parameters, and using performance metrics as the base scenarios for optimisation.

	Before	After
Positive	70	350
Negative	113	380
Total	183	730

Table 3.1 Images Before and after Image Agumentation

Model	Accuracy	Precision	Recall	F1-Score
DenseNet-121	92%	0.97	0.89	0.93
MobileNetV2	87%	0.97	0.81	0.88
VGGNet-19	69%	0.69	0.86	0.77
YOLOv8 Classification	88%	0.99	0.80	0.87
EfficientNet	73%	0.72	0.85	0.78
InceptionNet	97%	0.97	0.98	0.98
SVC	78%	0.79	0.83	0.81
Custom CNN	66%	0.67	0.81	0.73
ResNet 50 & 101	84%	0.83	0.83	0.83

Table 3.2 Results of Individual Models.

Chapter 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

A brief overview of the learning objectives set by the company for interns working on the project was:

- 1. Machine Learning Fundamentals:** Gain a solid understanding of fundamental concepts in machine learning, including supervised learning, classification, and model evaluation.
- 2. Deep Learning and CNNs:** Learn advanced techniques in deep learning, focusing on Convolutional Neural Networks (CNNs) and their applications in image classification tasks.
- 3. Image Pre-Processing Techniques:** Acquire knowledge of image preprocessing techniques such as data augmentation, normalisation, and feature extraction, essential for preparing datasets for machine learning models.
- 4. Model Development and Optimisation:** Develop skills in designing, implementing, and optimising CNN architectures for specific tasks like melanoma detection. Learn techniques for hyperparameter tuning and model evaluation to improve performance.
- 5. Experimentation and Analysis:** Gain experience in designing experiments, collecting and analysing data, and interpreting results. Develop critical thinking and problem-solving skills through iterative experimentation and analysis.
- 6. Project Management and Collaboration:** Enhance project management skills by coordinating tasks, meeting deadlines, and communicating effectively with team members. Learn the importance of collaboration and teamwork in achieving project objectives.
- 7. Ethical and Regulatory Awareness:** Understand the ethical implications of working with healthcare data and the importance of adhering to regulatory standards and guidelines. Learn best practices for protecting patient privacy and ensuring data security.

By focusing on these learning objectives, interns can develop a comprehensive skill set in machine learning, deep learning, image processing, and project management, while also gaining valuable insights into ethical considerations in healthcare technology.

Chapter 5

CHALLENGES AND SOLUTIONS

1. Variability in Image Quality:

- Challenge: Immunochromatographic card test images may exhibit variability in quality due to factors such as lighting conditions, device specifications, and user handling.
- Solution: Implement robust image preprocessing techniques, including brightness and exposure adjustments, to standardise image quality and enhance clarity. Additionally, conduct thorough quality control measures during image acquisition to ensure consistency.

2. Segmentation Accuracy:

- Challenge: YOLO v8 segmentation may encounter difficulties accurately delineating the regions of interest, particularly in cases of faint or overlapping test lines.
- Solution: Fine-tune the segmentation model parameters and incorporate post-processing techniques, such as contour analysis and morphological operations, to refine the segmentation results and improve accuracy.

3. Feature Extraction Complexity:

- Challenge: Extracting relevant features from preprocessed images, especially subtle variations in test line intensity and morphology, may pose challenges.
- Solution: advanced feature extraction algorithms tailored to the specific characteristics of immunochromatographic card test images. Consider employing deep learning-based feature extraction methods to capture nuanced patterns and relationships in the data.

4. Limited Training Data:

- Challenge: Obtaining a sufficient quantity of annotated immunochromatographic card test images for training the Inception Net model may be challenging, particularly for rare or specialised variants.
- Solution: Augment the available training data through techniques such as data synthesis, data augmentation, and transfer learning from related datasets. Additionally, collaborate

with healthcare institutions and research organisations to acquire diverse and representative image samples.

5. Model Generalisation and Robustness:

- Challenge: Ensuring the trained Inception Net model generalises well to unseen data and remains robust in real-world deployment scenarios is essential for reliable performance.
- Solution: Regularly validate the model on independent test datasets to assess generalisation performance. Employ techniques such as regularisation, dropout, and ensemble learning to enhance model robustness and mitigate overfitting. Conduct thorough validation and calibration procedures to verify performance consistency across different environments and conditions.

6. Regulatory Compliance and Ethical Considerations:

- Challenge: Adhering to regulatory requirements and ethical guidelines governing medical device development, data privacy, and patient confidentiality is crucial.
- Solution: Collaborate with regulatory experts and legal advisors to ensure compliance with relevant regulations and standards, such as FDA guidelines for medical imaging devices. Implement robust data security protocols, anonymization techniques, and informed consent procedures to safeguard patient privacy and confidentiality.

Chapter 6

ACHIEVEMENTS AND CONTRIBUTION

Throughout the development and implementation of the project, our team made significant achievements and contributions that have advanced the field of medical diagnostics and paved the way for improved detection of typhoid infection using immunochromatographic card tests. Below are the key achievements and contributions made by our team members:

1. Development of a Image Segmentation Model:

- One of the primary achievements of our project was the successful development of a robust image segmentation model based on YOLO v8 architecture. This model accurately delineates the regions of interest within immunochromatographic card test images, laying the foundation for subsequent analysis.

2. Implementation of Advanced Image Preprocessing Techniques:

- Our team implemented sophisticated image preprocessing techniques, including erosion, dilation, brightness, and exposure adjustments, to enhance the quality and clarity of the segmented regions. These preprocessing steps were instrumental in improving the visibility of test lines and facilitating more accurate feature extraction.

3. Integration of the Inception Net Model for Classification:

- We integrated the powerful Inception Net model for image classification, enabling the automated classification of immunochromatographic card test results as positive or negative for typhoid infection. This model leverages learned patterns and features to provide reliable assessments of test outcomes.

4. Optimisation for Real-Time Processing:

- Our efforts also focused on optimising the computational efficiency of the segmentation and classification models to enable real-time processing of test images. By minimising

computational complexity and leveraging hardware acceleration techniques, we achieved efficient performance suitable for deployment in resource-constrained environments.

5. Validation and Performance Assessment:

- We conducted rigorous validation and performance assessment procedures to ensure the accuracy and reliability of our diagnostic system. Thorough testing on independent validation datasets and quantitative evaluation using robust metrics enabled us to validate the efficacy of our approach and demonstrate its suitability for real-world applications.

6. Contribution to Healthcare Accessibility and Affordability:

- By developing an intelligent image processing solution for typhoid detection using immunochromatographic card tests, our project contributes to enhancing healthcare accessibility and affordability. Our automated diagnostic system streamlines the analysis process, reducing the dependence on manual interpretation and enabling faster and more reliable diagnoses.

7. Potential for Impact in Global Health Initiatives:

- Our project holds significant potential for impact in global health initiatives aimed at combating infectious diseases such as typhoid. The integration of advanced imaging and deep learning technologies into diagnostic workflows has the potential to revolutionise disease detection and surveillance, particularly in resource-limited settings.

8. Collaborative Team Effort:

- Finally, our achievements and contributions are a testament to the collaborative effort of our team members, each contributing their unique expertise and skills to the project. Through effective communication, coordination, and teamwork, we were able to overcome challenges, innovate solutions, and achieve our project objectives.

These achievements and contributions collectively represent a significant step forward in the advancement of medical diagnostics, with the potential to positively impact patient outcomes and public health on a global scale.

Chapter 7

REFLECTION AND ANALYSIS

Here's a reflection and analysis of my involvement in the "Intelligent Image Processing for Immunochromatographic Card Tests for Typhoid Detection" project:

My participation in the project was multifaceted, encompassing the development and implementation of image segmentation and preprocessing techniques crucial for accurate typhoid detection using immunochromatographic card tests. Through my contributions in model development, preprocessing pipeline design, and performance evaluation,

In my role as a developer of image segmentation models, I encountered various challenges and opportunities for learning. By refining parameters and optimising architectures, I honed my skills in fine-tuning models for specific application domains. One of the highlights of my involvement was the exploration and implementation of advanced image preprocessing techniques. From erosion and dilation operations to brightness and exposure adjustments, I learned how to tailor preprocessing strategies to enhance the quality and interpretability of input data. This experience emphasised the critical role of preprocessing in improving downstream analysis and classification accuracy.

Throughout the project, I demonstrated strong analytical skills and attention to detail in evaluating model performance and refining methodologies iteratively. By documenting experimental results and findings meticulously, I facilitated effective collaboration and knowledge sharing within the team. Moreover, the process of validation and performance assessment reinforced the importance of robust evaluation metrics and validation procedures in ensuring the reliability and efficacy of our diagnostic system.

Overall, my involvement in the project provided a valuable opportunity for learning, growth, and skill development in the fields of machine learning, image processing, and healthcare technology. By contributing to the development of intelligent image processing techniques for typhoid detection, I not only advanced the project's objectives but also gained a deeper understanding of the complexities and challenges inherent in applying AI solutions to real-world healthcare problems.

Chapter 8

RECOMMENDATIONS

To further advance "Intelligent Image Processing for Immunochromatographic Card Tests for Typhoid Detection," the following strategic recommendations are proposed:

1. Exploration of Ensemble Learning Techniques:

- Utilise ensemble learning techniques like bagging or boosting to enhance predictive accuracy and reduce model variance. By combining multiple segmentation and classification models, the project can improve typhoid detection reliability.

2. Incorporation of Transfer Learning:

- Implement transfer learning with pre-trained models like Inception Net on diverse datasets to expedite model training and boost generalisation capabilities, especially in scenarios with limited annotated data.

3. Integration of Explainable AI Techniques:

- Incorporate explainable AI methods such as attention mechanisms or saliency mapping to improve the interpretability of segmentation and classification models. This facilitates validation by healthcare professionals, fostering trust in the diagnostic system.

4. Establishment of Continuous Model Monitoring and Updating Protocols:

- Set up protocols for ongoing model monitoring and updates to maintain long-term efficacy. Regular assessment and adaptation to evolving clinical practices ensure sustained accuracy and reliability in typhoid detection outcomes.

By implementing these strategic recommendations, the project can further optimize its intelligent image processing solution for immunochromatographic card test analysis, ultimately maximizing its impact on healthcare diagnostics and patient care.

Chapter 9

CONCLUSION

In conclusion, this research has demonstrated the remarkable potential of intelligent image processing using deep learning models in the context of interpreting immunochromatographic test (ICT) results. The utilisation of the YOLOv8 model for image segmentation yielded an impressive Mean Average Precision (mAP) score of 92, showcasing its efficacy in precisely delineating regions of interest within ICT card images. Furthermore, our findings highlight InceptionNet as the optimal choice for classification, attaining an outstanding accuracy rate of 97% when compared to other prominent models such as DenseNet-121, MobileNetV2, VGGNet-19, EfficientNet, SVC, Custom CNN, ResNet 50, and ResNet 101.

Looking ahead, it becomes crucial for laboratories to consider investing in state-of-the-art automated systems that not only enhance accuracy but also significantly improve the overall efficiency of the diagnostic process. These advances in technology hold the potential to expedite diagnoses and streamline workflows. Nonetheless, it is imperative to underscore that human oversight remains indispensable. Results generated by automated systems should always be subject to verification by trained personnel to ensure their clinical relevance and accuracy. judgement

Striking a balance between cutting-edge technology and expert human judgment will undoubtedly pave the way for more accurate, efficient, and reliable diagnostic practices in the realm of medical testing. Furthermore, it is essential to acknowledge that this research is an initial step in harnessing the potential of artificial intelligence in the context of ICT result interpretation.

This study lays the foundation for further exploration and refinement. The field of medical diagnostics and image processing is dynamic, with ongoing advancements and innovations. Consequently, this study remains adaptable for future refinements, holding the potential to further boost the precision and efficiency of diagnostic processes in the foreseeable future.

Chapter 10

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